

Business Analytics Case Study

Bank Marketing Dataset: Predicting Term Deposit Subscription

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1. Introduction

Direct marketing campaigns — where a bank's staff contact customers by phone to offer a product — remain one of the most common ways banks sell term deposits. However, these campaigns are expensive and time-consuming, and only a fraction of customers contacted ultimately subscribe. This report analyzes historical direct-marketing call data from a Portuguese retail bank to understand what separates customers who subscribe to a term deposit from those who do not, and to translate that understanding into concrete recommendations that improve campaign efficiency.

2. Business Problem Statement

The bank's marketing team currently contacts a broad base of customers with limited targeting, resulting in a large number of calls that do not convert. The core business questions this study addresses are:

- Why do some customers subscribe to term deposits while others do not?
- Which customer groups are more likely to respond positively to marketing campaigns?

Answering these questions allows the bank to prioritize high-probability customers, reduce wasted contact effort, and increase the overall conversion rate of future campaigns.

3. Dataset Description

The dataset contains 11,162 customer records from past telemarketing campaigns, with 17 columns covering customer demographics, financial standing, and campaign contact history. The target variable is deposit (yes/no), indicating whether the customer subscribed to a term deposit.

Attribute	Detail
Rows	11,162 customer records
Columns	17 (demographic, financial, and campaign attributes)
Target variable	deposit (yes / no)
Missing values	0 nulls; four categorical fields (job, education, contact, poutcome) contain an 'unknown' category used as a placeholder
Class balance	52.6% did not subscribe, 47.4% subscribed — a well-balanced dataset

Key fields used in the analysis include age, job, marital status, education, balance (account balance), housing/personal loan status, contact method, month and day of last contact, call duration, number of campaign contacts, days since previous contact (pdays), number of previous contacts, and the outcome of the previous campaign (poutcome).

4. Data Analysis Process

The analysis followed a structured exploratory data analysis (EDA) process: (1) inspecting dataset shape, data types, and missing values; (2) computing summary statistics for numeric fields; (3) visualizing distributions of key variables; (4) cross-tabulating categorical variables against the subscription outcome to estimate conversion rates

per segment; and (5) building a correlation heatmap across numeric fields to check linear relationships with subscription. Python (Pandas, Seaborn, Matplotlib) was used throughout.

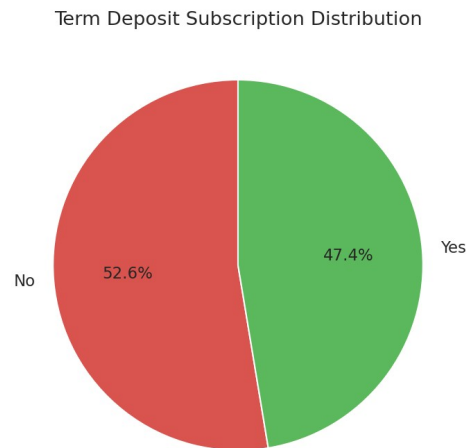


Figure 1: Overall subscription outcome is close to balanced (47.4% yes).

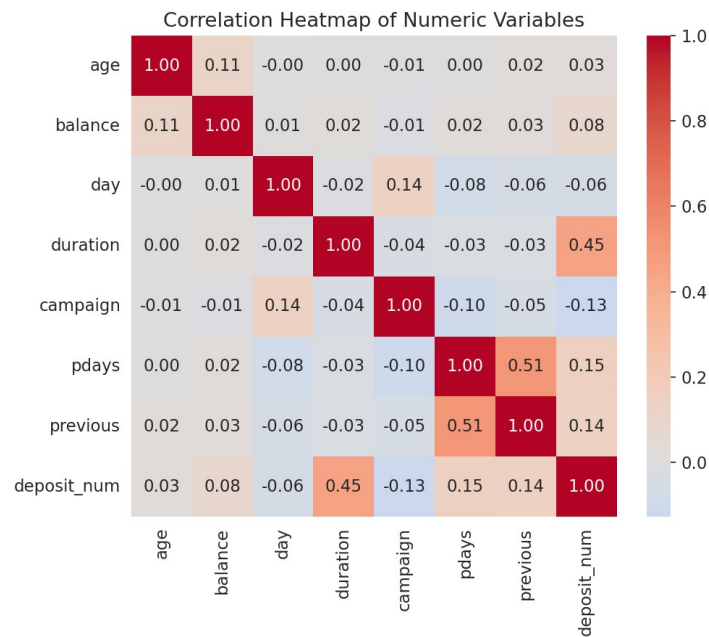


Figure 2: Correlation heatmap — call duration shows the strongest positive relationship with subscription.

5. Key Findings & Key Drivers

5.1 Call Duration Is the Single Strongest Driver

Customers who subscribed had an average call duration of 537 seconds (~9 minutes), versus just 223 seconds (~3.7 minutes) for those who did not. Duration correlates with the outcome at 0.45 — by far the strongest numeric relationship in the dataset. Longer, more engaged conversations are strongly associated with conversion.

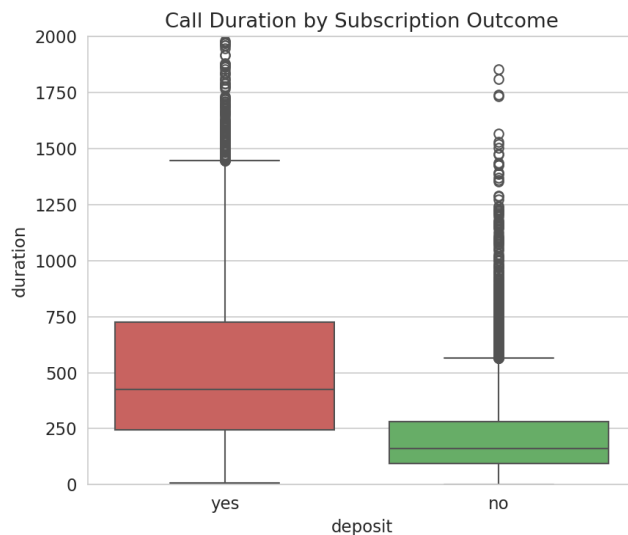


Figure 3: Call duration is markedly higher for customers who subscribed.

5.2 Prior Campaign Success Predicts Future Success

Customers whose previous campaign outcome was a 'success' subscribed 91.3% of the time in this campaign — far above the 40.7% baseline rate for customers with no prior contact history. Even a 'failure' in a prior campaign (50.3%) outperforms cold, first-time contacts.

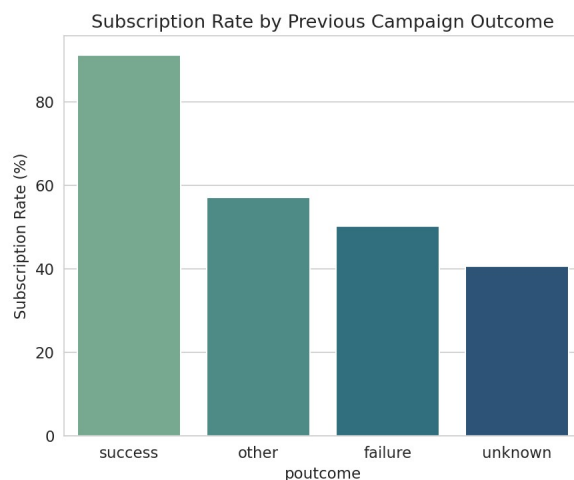


Figure 4: Subscription rate jumps sharply for customers with a prior successful campaign outcome.

5.3 Job, Age, and Education Segment Customers Clearly

Students (74.7%) and retirees (66.3%) subscribe at far higher rates than blue-collar workers (36.4%) or entrepreneurs (37.5%). Age tells a similar story: customers aged 65+ (80.4%) and 18-25 (71.1%) — the two ends of the working-age spectrum — subscribe well above the 47.4% average, while the 36-55 'prime working age' band

lags at around 41-42%. Education also matters: tertiary-educated customers subscribe at 54.1% versus 39.4% for primary education.

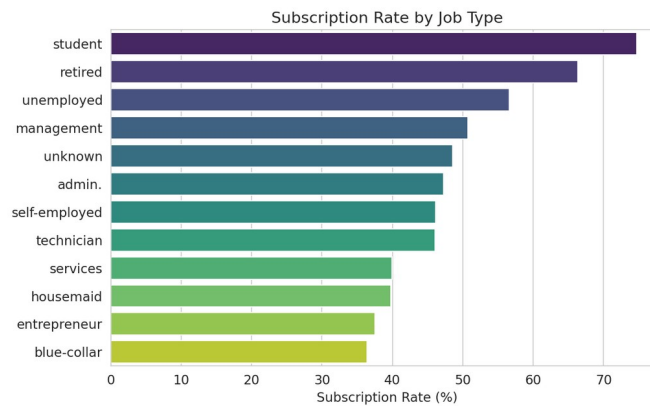


Figure 5: Students and retirees are the strongest-converting job segments.

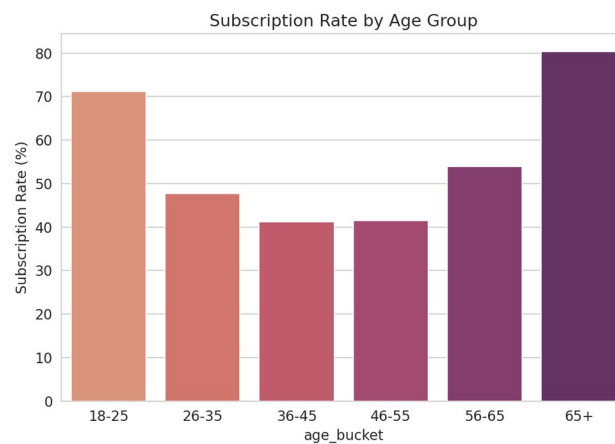


Figure 6: Subscription rate is highest among the youngest and oldest age groups.

5.4 Contact Method and Timing Matter

Customers reached by cellular phone subscribed at 54.3%, compared to just 22.6% for contacts logged as 'unknown' method. Seasonality is also pronounced: December, March, September, and October show subscription rates above 82%, while May — the busiest calling month by volume — has the lowest conversion rate at just 32.8%, suggesting the bank may be over-investing effort in a low-yield period.

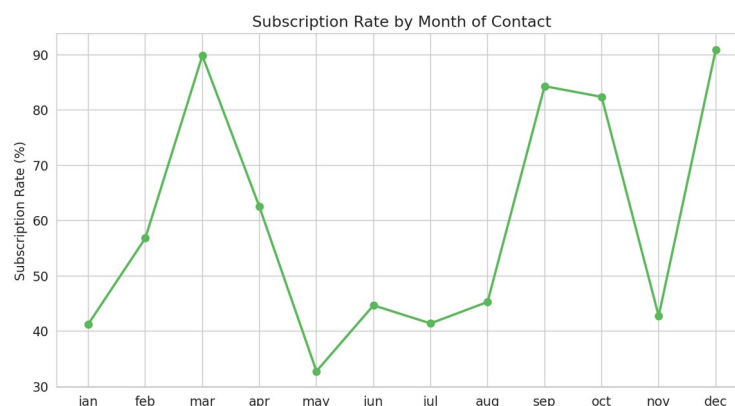


Figure 7: Subscription rate varies sharply by contact month; May, the highest-volume month, converts worst.

5.5 Financial Standing and Existing Debt Reduce Conversion

Customers with no housing loan subscribe at 57.0% versus 36.6% for those with an active housing loan; the same pattern holds for personal loans (49.5% vs 33.2%). Higher account balances are also associated with higher subscription: customers in the top balance quartile subscribe at 57.5%, compared to 36.0% in the bottom quartile.

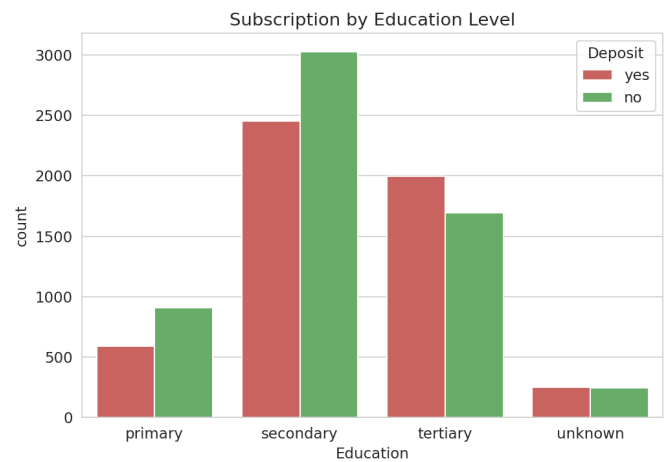


Figure 8: Subscription rises with education level across all deposit outcomes shown.

6. Business Insights

- Engagement quality (call duration) outweighs contact volume — a bank rep who keeps a customer on the phone longer is far more likely to close, so scripts and rep training should prioritize sustaining quality conversation over rushing through call lists.
- Warm leads (previous campaign contacts, especially past successes) convert dramatically better than cold outreach — the current campaign appears to under-leverage this history.
- Life-stage segments at the edges — students and retirees — are under-served relative to their conversion potential, while heavy calling volume is directed at prime-working-age blue-collar segments that convert far less often.
- Debt-free, higher-balance customers are financially better positioned to commit funds to a term deposit, explaining their higher conversion; indebted customers likely have less discretionary balance to lock away.
- Calling volume and calling effectiveness are misaligned by month — May absorbs the most calls but returns the lowest conversion, representing a clear efficiency risk.
- Segments requiring attention: blue-collar and entrepreneur job categories, customers with active housing/personal loans, and the 36-55 age band — all convert below the dataset average and represent either a targeting risk or an opportunity for a different offer/approach.

7. Recommendations

- 1. Prioritize call-list ordering by prior campaign outcome: contact customers with a previous 'success' or 'other' outcome first, since they convert at 2x the base rate.
- 2. Reallocate campaign volume away from May toward higher-converting months (Dec, Mar, Sep, Oct) where feasible, or redesign the May offer/script given its consistently low yield.
- 3. Build targeted messaging for students and retirees (e.g., education/starter savings framing for students, stable-income/retirement-income framing for retirees) given their outsized conversion rates, and test a differentiated offer for blue-collar and entrepreneur segments to lift their lagging conversion.
- 4. Invest in call-quality training rather than call-volume targets — coach reps to extend engagement (needs-based questions, addressing objections) since duration is the strongest predictor of a sale.
- 5. Prefer cellular contact over methods logged as 'unknown', and prioritize customers with no active housing/personal loan and higher account balances, who are both more reachable and more likely to convert.

8. Conclusion

This analysis shows that term deposit subscription is not random: it is strongly predictable from call engagement, campaign history, life stage, and financial standing. By shifting from broad, volume-driven outreach toward a targeted strategy that prioritizes warm leads, favorable months, and receptive demographic segments, the bank can meaningfully raise its conversion rate while reducing the number of low-yield calls its team needs to make. The recommendations above translate these data-driven patterns into concrete, actionable next steps for the marketing team.